

OULU AP, Finland

WMO: 028750

Lat: 64.926N

Lon: 25.373E

Elev: 14

StdP: 101.15

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-24.0	-30.0	0.2	-27.1	-26.4	0.3	-23.8	10.4	-1.4	9.3	-2.7	1.7	130	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	25.8	17.9	23.8	16.7	21.9	15.9	19.4	23.7	18.0	21.8	16.9	20.5	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	12.8	21.3	16.5	11.7	19.9	15.2	10.8	18.7	55.0	23.3	51.0	21.8	47.7	20.6	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.0	7.9	7.0	-29.4	28.4	4.2	2.5	-32.4	30.3	-34.9	31.7	-37.3	33.1	-40.3	35.0		
(5)			-29.5	20.6	4.2	1.8	-32.5	21.9	-35.0	22.9	-37.3	23.9	-40.4	25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-8.9	-8.9	-4.7	1.7	7.8	13.4	16.8	14.6	9.6	3.4	-1.7	-5.6
	DBStd	10.17	7.45	7.38	5.28	3.66	4.39	3.51	3.03	3.11	3.39	4.07	5.08	7.02
	HDD10.0	3019	586	529	456	249	97	8	0	2	48	206	352	485
	HDD18.3	5560	844	763	714	499	327	154	68	121	262	463	602	743
	CDD10.0	532	0	0	0	1	29	109	209	146	36	1	0	0
	CDD18.3	31	0	0	0	0	1	5	19	6	0	0	0	0
	CDH23.3	212	0	0	0	0	12	43	119	37	0	0	0	0
	CDH26.7	37	0	0	0	0	1	9	22	4	0	0	0	0
(14) Wind	WSAvg	3.3	3.3	3.4	3.5	3.2	3.6	3.5	3.1	3.0	3.2	3.3	3.4	3.5
(15) Precipitation	PrecAvg	442	28	22	23	20	33	44	60	61	47	41	33	29
	PrecMax	632	72	53	48	55	92	102	178	123	106	91	71	62
	PrecMin	271	6	0	5	5	1	4	0	0	8	0	8	7
	PrecStd	81	14	12	12	13	20	24	37	26	22	22	15	12
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	5.0	7.0	15.6	25.0	28.0	29.2	26.8	20.9	13.1	9.0	5.2
		MCWB	2.4	3.0	3.5	8.9	14.7	17.7	19.4	18.9	15.0	10.8	7.6	4.5
	2%	DB	2.2	3.0	4.9	12.0	21.0	24.0	26.2	24.0	18.1	11.4	7.2	4.0
		MCWB	1.4	1.8	2.3	7.1	13.1	15.8	18.8	17.1	13.7	9.8	6.5	3.3
	5%	DB	1.2	1.5	3.2	9.3	17.9	21.2	24.2	22.0	16.2	10.1	5.8	3.0
		MCWB	0.7	0.8	1.3	5.3	11.3	14.3	17.6	16.2	13.0	8.8	5.1	2.2
	10%	DB	0.2	0.2	2.1	7.2	15.1	19.6	22.6	20.2	14.9	9.0	4.2	1.9
		MCWB	-0.1	-0.3	0.7	4.0	9.6	13.7	16.9	15.5	12.4	7.9	3.7	1.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	3.1	4.0	10.0	15.5	19.0	21.3	20.3	15.9	11.7	8.0	4.8
		MCDB	3.6	4.3	6.3	14.0	22.6	24.3	25.6	24.9	19.0	12.9	8.5	5.0
	2%	WB	1.6	1.7	2.6	7.3	13.6	17.1	20.0	18.6	14.5	10.2	6.5	3.2
		MCDB	2.0	2.5	4.2	11.1	19.7	21.5	24.5	22.2	17.1	11.0	7.0	3.6
	5%	WB	0.8	0.8	1.6	5.6	11.8	15.7	18.7	17.4	13.5	9.0	5.0	2.1
		MCDB	1.2	1.2	3.0	8.6	16.9	19.9	22.8	20.5	15.8	9.9	5.4	2.5
	10%	WB	-0.1	-0.1	0.7	4.3	10.3	14.4	17.6	16.2	12.7	7.8	3.7	1.2
		MCDB	0.3	0.3	1.7	6.8	14.1	18.5	21.3	19.2	14.7	8.7	4.1	1.6
(35) Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	7.8	7.9	8.4	7.9	7.9	8.0	7.0	4.9	4.5	5.5
		MCDBR	5.2	5.5	7.3	11.3	13.0	11.4	11.0	10.9	9.1	5.4	4.9	4.2
		MCWBR	4.9	4.9	5.6	7.1	6.6	5.4	5.2	5.5	5.7	4.3	4.5	3.8
	5% WB	MCDBR	5.2	5.2	6.0	10.3	11.6	9.7	9.7	9.0	8.0	5.2	4.4	4.2
		MCWBR	4.9	4.8	4.9	6.8	6.5	5.3	5.2	5.4	5.4	4.4	4.2	3.9
(40) Clear-Sky Solar Irradiance	taub		0.209	0.278	0.308	0.346	0.347	0.350	0.370	0.353	0.330	0.307	0.344	0.355
	taud		1.952	2.110	2.163	2.247	2.428	2.468	2.435	2.473	2.519	2.359	2.554	2.610
	Ebn at Noon		300	599	758	818	861	865	832	813	756	582	276	120
	Edh at Noon		23	60	89	105	97	96	96	85	65	48	21	12
(44) All-Sky Solar Radiation	RadAvg		0.11	0.62	2.10	3.78	4.96	5.55	5.12	3.80	2.09	0.76	0.16	0.03
	RadStd		0.01	0.12	0.21	0.30	0.47	0.45	0.46	0.36	0.21	0.12	0.03	0.00

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page